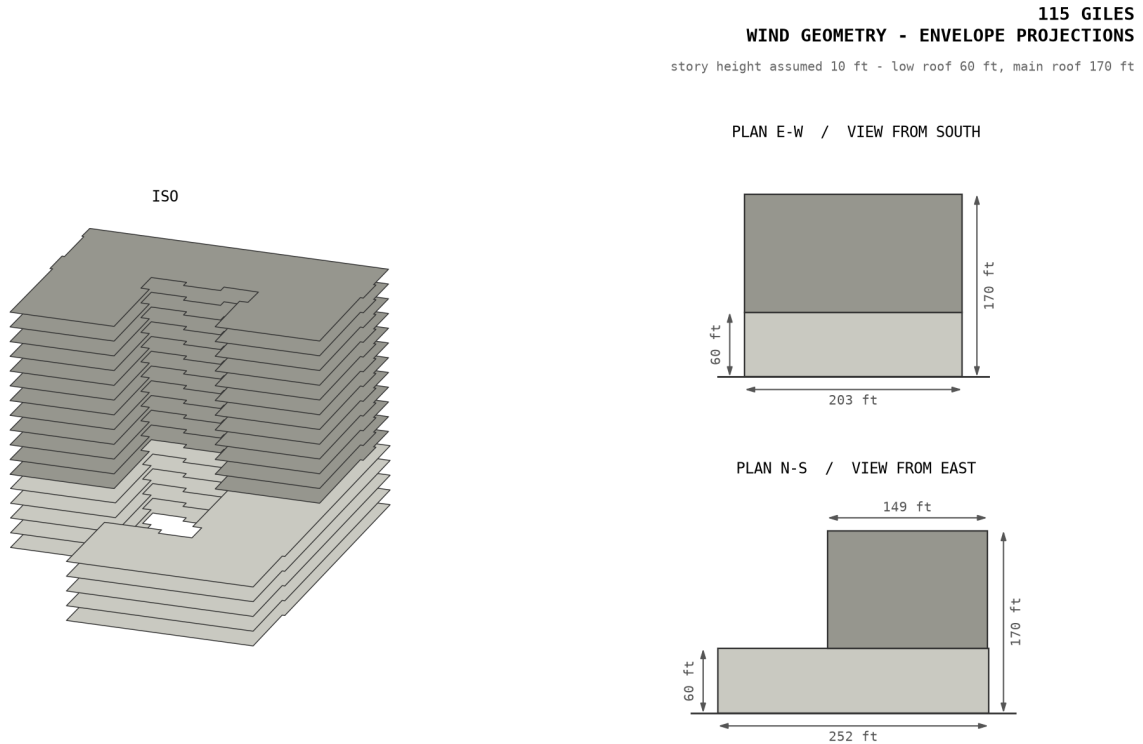


1 WIND

1.1 GEOMETRY



1.2 VELOCITY PRESSURE

$$q_z = 0.00256 K_z K_{zt} K_e K_d V^2 = 0.00256(1.09)(1.0)(1.0)(0.85)(115)^2 = 31.4 \text{ psf} \quad (1)$$

K_z exposure coefficient, Exp B at $z = h = 170$ ft (ASCE 7-22)
 K_{zt} topographic factor
 K_e ground elevation factor
 K_d directionality factor
 V basic wind speed, 115 mph, Risk Cat II, Jersey City NJ

Adopt $q_z = 35$ psf uniform over full height ($\approx 11\%$ over q_h).

1.3 NET PRESSURE (WINDWARD + LEEWARD)

$$p = q_z G (C_{p,ww} + |C_{p,lw}|) = 35(0.85)(0.8 + 0.5) = 38.7 \text{ psf} \quad (2)$$

G gust-effect factor
 $C_{p,ww}$ windward wall pressure coefficient, +0.8
 $C_{p,lw}$ leeward wall pressure coefficient, -0.5 ($L/B \leq 1$; N-S wind $L/B \approx 1.25 \rightarrow -0.45$, use -0.5)

Adopt $p = 40$ psf net on projected area, both directions.

1.4 BASE SHEAR

$$V_x = p A_x = 40 (252 \cdot 60 + 149 \cdot 110) / 1000 = 1,260 \text{ k} \quad (3)$$

$$V_y = p A_y = 40 (203 \cdot 170) / 1000 = 1,380 \text{ k} \quad (4)$$

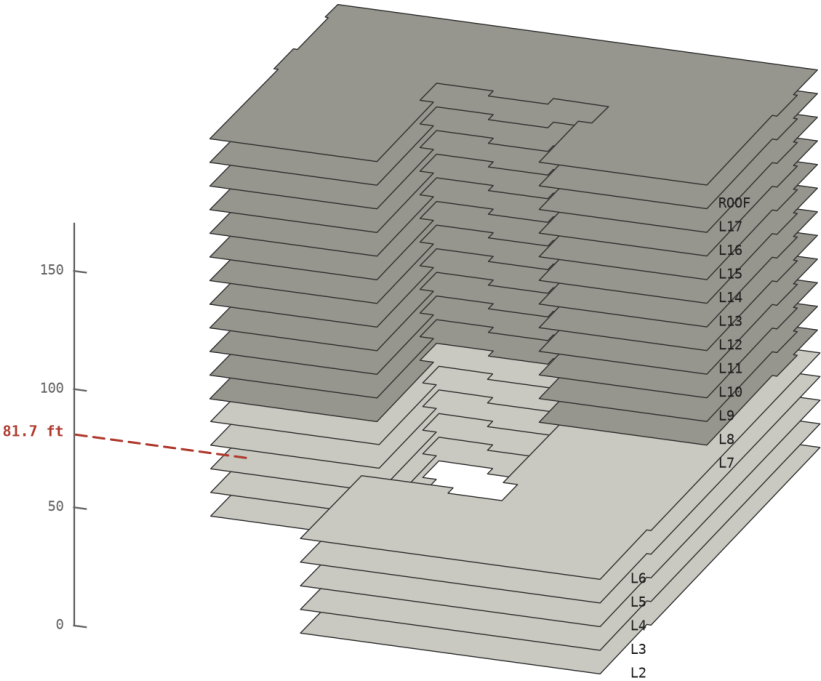
A_x projected area for wind along X (E-W), N-S elevation profile
 A_y projected area for wind along Y (N-S), E-W elevation profile

2 EQ

2.1 GEOMETRY

LOWER	35,765 sf x 5 (L2-L6)	= 178,823 sf
UPPER	23,141 sf x 12 (L7-R00F)	= 277,691 sf
TOTAL		= 456,514 sf
MEAN HT	$\text{sum}(A \cdot z) / \text{sum}(A)$	= 81.7 ft

115 GILES
CRUDE SLAB MASSING
story height assumed 10 ft



2.2 LOADS

$$w = w_{slab} + w_{SDL} = 100 + 35 = 135 \text{ psf} \tag{5}$$

w_{slab} 8 in slab self-weight, 100 psf, all floors
 w_{SDL} superimposed dead load, 35 psf, all floors

$$W = w \Sigma A = 0.135(456,514) = 61,629 \text{ k} \tag{6}$$

W effective seismic weight
 ΣA total slab area above base, per 2.1

2.3 HAZARD / SITE

Case study: ASCE 7-16 site-coefficient chain vs ASCE 7-22 direct multi-period, both Site Class D assumed (no geotech).

ASCE 7-16

$$\begin{aligned} S_{MS} &= F_a S_S = 1.58(0.28) = 0.44 \\ S_{M1} &= F_v S_1 = 2.40(0.051) = 0.12 \\ S_{DS} &= \frac{2}{3} S_{MS} = 0.30 \\ S_{D1} &= \frac{2}{3} S_{M1} = 0.082 \end{aligned}$$

S_S, S_1 mapped MCE_R accelerations, 0.28 / 0.051
 F_a, F_v site coefficients (7-16 Tables 11.4-1/2)
 S_{MS}, S_{M1} site-adjusted MCE_R ; 7-22 values direct from ASCE 7 Hazard Tool
 S_{DS}, S_{D1} design accelerations

ASCE 7-22

$$\begin{aligned} S_{MS} &= 0.30 \text{ (tool, direct)} \\ S_{M1} &= 0.10 \text{ (tool, direct)} \\ S_{DS} &= \frac{2}{3} S_{MS} = 0.20 \\ S_{D1} &= \frac{2}{3} S_{M1} = 0.067 \end{aligned}$$

Risk Cat II, $I_e = 1.0$. SDC B both cases.

2.4 SYSTEM / PERIOD

Ordinary reinforced concrete shear walls, bearing wall system, $R = 4$.

$$T_a = C_t h_n^x = 0.02(170)^{0.75} = 0.94 \text{ s} \quad (7)$$

R response modification coefficient
 C_t, x approximate-period coefficient / exponent, all other systems
 h_n structural height above base, 170 ft

2.5 BASE SHEAR (ELF)

$$C_s = \frac{S_{DS}}{R/I_e} \quad C_{s,max} = \frac{S_{D1}}{T(R/I_e)} \quad C_{s,min} = 0.044 S_{DS} I_e \geq 0.01 \quad V = C_s W \quad (8)$$

C_s seismic response coefficient, bounded by $C_{s,max}$ / $C_{s,min}$
 T period used in direction considered, $T = T_a = 0.94 \text{ s}$

ASCE 7-16

$$\begin{aligned} C_s &= 0.30/4 = 0.075 \\ C_{s,max} &= \frac{0.082}{0.94(4)} = 0.0218 \leftarrow \\ C_{s,min} &= 0.044(0.30) = 0.0132 \\ V &= 0.0218(61,629) = \boxed{1,344 \text{ k}} \end{aligned}$$

ASCE 7-22

$$\begin{aligned} C_s &= 0.20/4 = 0.050 \\ C_{s,max} &= \frac{0.067}{0.94(4)} = 0.0178 \leftarrow \\ C_{s,min} &= 0.01 \text{ (floor)} \\ V &= 0.0178(61,629) = \boxed{1,097 \text{ k}} \end{aligned}$$

7-16 governs, +23%. $C_{s,max}$ controls both cases.